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Sugar rationing during the first 1000 days of life and lifelong risk of heart failure

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Elevated sugar intake has been linked to poor cardiovascular health, but the effects of early-life sugar restriction on the lifelong risk of heart failure remain unclear. Using the end of sugar rationing in the United Kingdom in 1953 as a natural experiment, we classify participants in the UK Biobank by birth timing to define early-life exposure to sugar rationing. Here we show that individuals exposed to early-life sugar rationing exhibit an approximately 14% lower risk of heart failure and develop the condition about 2.6 years later than those unexposed. Longer exposure is associated with stronger effects, and population-level estimates suggest that around 4–5% of heart failure cases may be attributable to the absence of early-life sugar restriction. Genetic susceptibility does not modify these associations but acts additively. These findings underscore the long-term importance of early-life sugar exposure for heart failure, and further studies are warranted.

Heart failure (HF) is characterised as a syndrome comprising symptoms or signs resulting from cardiac abnormalities and is confirmed by elevated natriuretic peptides or evidence of congestion[1]. Estimates from the Global Burden of Disease Study indicate that approximately 55.5 million individuals globally were affected by HF in 2021[2,3]. Notably, HF is associated with considerable health challenges, including reduced quality of life, recurrent hospitalisations, rising healthcare expenses and an increased risk of premature death[4].

High sugar intake is a potential risk factor for HF. Previous prospective studies in adults have demonstrated a strong correlation between increased consumption of sugar-sweetened beverages (SSBs) or artificially sweetened beverages (ASBs) and the incidence of HF[5–7]. Although restricting sugar intake during childhood or early life had been considered to mitigate cardiometabolic risk factors such as hypertension, diabetes mellitus (DM), and hypercholesterolemia[8–11], evidence inadequate to determine whether restricting sugar intake during pregnancy or early life reduced the risk of HF later in life.

In July 1942, the United Kingdom (UK) implemented rationing for sugar and sweets as a component of a 14-year food rationing program during World War II[8,12]. During this period, sugar allowances for all individuals, including pregnant women and children, aligned with contemporary dietary guidelines: less than $40 \times g$ per day for adults and under $15 \times g$ for children, and no intake recommended for those under two years of age[8,13,14]. Gracner et al.[8] indicated that the cessation of rationing in September 1953 nearly doubled the consumption of sugar and sweets, which increased from $41 \times g$ in the first quarter of

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1953 to approximately 80 × *g* by the third quarter of 1954 (Fig. 1a); this increase accounted for at least 77% of the total increase in calorie intake, whereas the intake of other foods and nutrients remained largely unchanged; employing this natural experiment, they compared individuals conceived immediately before and after the cessation of rationing, reflecting cohorts with likely presumably divergent early-life sugar exposure, as the timing of birth entirely determined an individual had experienced rationing; their findings suggested that restricted sugar intake during early development was associated with approximately 35% and 20% reductions in the risks of DM and hypertension, respectively; additionally, disease onset was delayed by four years for DM and two years for hypertension[8].

This study aims to further investigate the influence of the first 1000 days post-conception, encompassing gestation to age 2, which is a crucial phase where an unhealthy diet may negatively impact adult health[15,16], on the risk of HF across the life course.

Results

Baseline characteristics of the participants

This study included 61,193 participants from the UK Biobank (UKB), and the mean recruitment age was 54.61 years (standard deviation [SD], 1.56) (Table 1). Approximately 56.21% were females, and 22,456 did not experience sugar rationing. A higher proportion of participants subjected to sugar rationing were White and born in England and had parents diagnosed with cardiovascular disease (CVD). A small proportion had parents with DM or hypertension. Although chi-squared tests indicated statistical significance between the two groups, the effect sizes were small and probably not substantial (Cramér's $V < 0.1$). No significant differences were noted between the two groups in terms of sex and polygenic risk score (PRS) tertiles. The percentage of participants who developed HF was slightly higher in the sugar rationing group than in the non-exposed group (2.66% vs 2.33%), but the age at HF diagnosis was considerably delayed in the exposed group (63.23 vs 60.61 years).

Sugar rationing and HF

Figure 2 illustrates the cumulative incidence of HF throughout the follow-up period. Landmark analysis indicated no significant difference between the groups in HF incidence before age 45 ($P = 0.767$). However, after age 45, the incidence of HF was significantly lower in participants exposed to sugar rationing ($P = 0.010$).

Table 2 illustrates that participants exposed to sugar rationing exhibited a significantly lower risk of developing HF than those who were not exposed (Cox regression, hazard ratio [HR] = 0.87, 95% confidence interval [CI]: 0.78–0.97; flexible parametric survival models [FPSM], HR = 0.86, 95% CI: 0.77–0.95). The outcomes remained consistent after controlling for covariates. Population attributable fractions (PAF) analysis indicated that undergoing sugar rationing in the initial 1000 days of life may have averted roughly 4%–5% of HF events. Compared to those who were never rationed + born 21 months after rationing, individuals exposed to sugar rationing for a longer duration had a lower risk of developing HF (P for trend <0.05) (Table 2).

Additional and sensitivity analysis

We further subdivided the duration of rationing into six-month intervals, yielding similar conclusions (Supplementary Table 1). Upon controlling for covariates, participants exposed to sugar rationing exhibited a delay of 2.57 years in the onset age of HF relative to those not exposed. The delay increased with the length of sugar rationing, reaching up to 4.50 years (Supplementary Table 2).

No significant multiplicative interaction was found between genetic susceptibility and sugar rationing, indicating that genetic susceptibility did not significantly alter the association between sugar rationing and HF. A notable additive interaction was observed (Table 3), exhibiting a positive synergistic effect (the relative excess risk due to interaction [RERI] or the attributable proportion due to interaction [AP] > 0, $P < 0.05$). Individuals with high genetic susceptibility who were not exposed to sugar rationing exhibited an increased risk of developing HF.

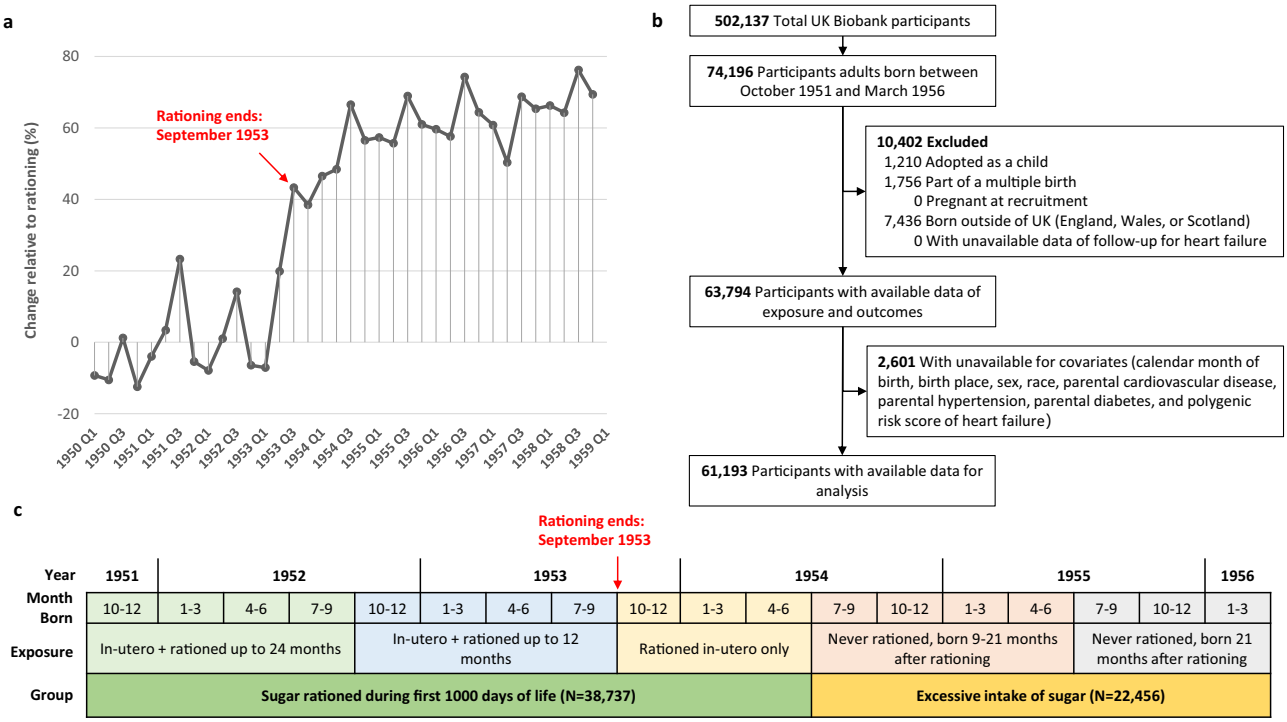

Fig. 1 | Study design. a Percent change in sugar intake relative to the rationing period (1950 Q1–1953 Q2). **b** enrollment and exclusion flowchart for the UK Biobank. **c** Timeline of sugar rationing exposure. Source data are provided as a Source Data file.

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Table 1 | Characteristics of survey participants born between October 1951 and March 1956

Characteristic	Overall *N* = 61,193	Non-rationed *N* = 22,456	Rationed *N* = 38,737	*P* value	Effect size
Age at recruitment, years				<0.001	−2.03
Mean (SD)	54.61 (1.56)	53.17 (1.00)	55.44 (1.18)		
Sex, *n* (%)				0.49	0.002
Female	34,398 (56.21)	12,664 (56.39)	21,734 (56.11)		
Male	26,795 (43.79)	9792 (43.61)	17,003 (43.89)		
Race, *n* (%)				<0.001	0.01
Others	432 (0.71)	195 (0.87)	237 (0.61)		
White	60,761 (99.29)	22,261 (99.13)	38,500 (99.39)		
Place of birth, *n* (%)				0.002	0.01
England	52,399 (85.63)	19,127 (85.18)	33,272 (85.89)		
Scotland	5736 (9.37)	2227 (9.92)	3509 (9.06)		
Wales	3058 (5.00)	1102 (4.91)	1956 (5.05)		
Parents diagnosed with CVD, *n* (%)				<0.001	0.02
No	25,472 (41.63)	9633 (42.90)	15,839 (40.89)		
Yes	35,721 (58.37)	12,823 (57.10)	22,898 (59.11)		
Parents diagnosed with DM, *n* (%)				<0.001	0.01
No	49,753 (81.31)	18,102 (80.61)	31,651 (81.71)		
Yes	11,440 (18.69)	4354 (19.39)	7086 (18.29)		
Parents diagnosed with hypertension, *n* (%)				<0.001	0.03
No	32,704 (53.44)	11,585 (51.59)	21,119 (54.52)		
Yes	28,489 (46.56)	10,871 (48.41)	17,618 (45.48)		
PRS tertile, *n* (%)				0.23	0.01
Low risk	20,446 (33.41)	7408 (32.99)	13,038 (33.66)		
Intermedia	20,251 (33.09)	7468 (33.26)	12,783 (33.00)		
High risk	20,496 (33.49)	7580 (33.75)	12,916 (33.34)		
Heart failure, *n* (%)				0.013	0.01
No	59,640 (97.46)	21,933 (97.67)	37,707 (97.34)		
Yes	1553 (2.54)	523 (2.33)	1030 (2.66)		
Age diagnosed with heart failure, years				<0.001	−0.46
Mean (SD)	62.35 (5.82)	60.61 (5.54)	63.23 (5.76)		

CVD cardiovascular disease, *DM* diabetes mellitus, *PRS* polygenic risk score, *SD* standard deviation. Group differences for continuous and categorical variables were assessed using the *t*-test and Pearson's chi-squared test, with all tests being two-sided. Effect sizes were measured using Cohen's *d* for continuous variables and Cramér's *V* for categorical variables.

In sensitivity analyses utilising the parametric proportional hazards model grounded in the Gompertz distribution, we found that individuals exposed to sugar rationing exhibited an 18% reduction in the risk of HF. The reduction reached nearly 30% as the duration of rationing increased (Supplementary Table 3). Upon the modification of the follow-up initiation point and the inclusion of additional confounding factors related to HF, the findings remained consistent. Even when individuals had reached middle age, early-life exposure to sugar rationing was still linked to reduced risk of HF in later life (Supplementary Tables 4 and Supplementary Fig. 1).

Discussion

We employed a natural experiment in the UK, revealing that sugar rationing during the initial 1000 days of life reduced the lifelong risk of HF and delayed its onset, especially in middle age. The effects of sugar rationing amplified with prolonged periods of restricted sugar intake. A notable positive synergistic effect was observed between genetic susceptibility and early-life sugar rationing.

Prior research has emphasised the potential risk of high sugar intake for HF. Zhang et al. conducted a study involving 209,829 adults from the UKB and discovered that compared with nonconsumers, individuals who consumed over 2 L per week of SSBs or ASBs exhibited a 22% and 30% heightened risk of developing HF, respectively, during a median follow-up of 9.9 years[7]. Rahman et al. conducted a study involving 42,400 males aged 45–79 years, revealing that individuals who consumed two or more servings of SSBs daily exhibited a 23% increased risk of developing HF compared with nonconsumers[5]. Yang et al. examined 109,034 females involved in the Women's Health Initiative and discovered that females who consumed one or more servings of SSBs daily exhibited a 35% higher risk of developing HF compared with those who consumed fewer than one serving of SSB weekly[6].

Evidence demonstrating how elevated sugar consumption in early life contributes to the onset of HF is scarce. Nevertheless, numerous studies have investigated intermediate outcomes[17]. Gracner et al. observed the protective effects of early-life sugar restriction against type 2 DM and hypertension, utilising a study design akin to that of the current research[8]. Leermakers et al. conducted a study involving 1316 Dutch children and discovered that increased consumption of SSBs at 13 months correlated with elevated cardiometabolic risk factor scores at 6 years of age. This score is a continuous measure derived from body fat percentage, blood pressure and insulin, high-density lipoprotein (HDL) cholesterol and triglyceride levels[9]. Numerous additional studies have recognised the benefits of sugar restriction for cardiometabolic health in children and adolescents, correlating with elevated HDL levels, diminished obesity rates, decreased triglycerides, total cholesterol, insulin resistance and systolic blood pressure[10,11,18–20]. Additional mechanistic hypotheses, including the developmental origins hypothesis, propose that modified long-term disease risk is initially

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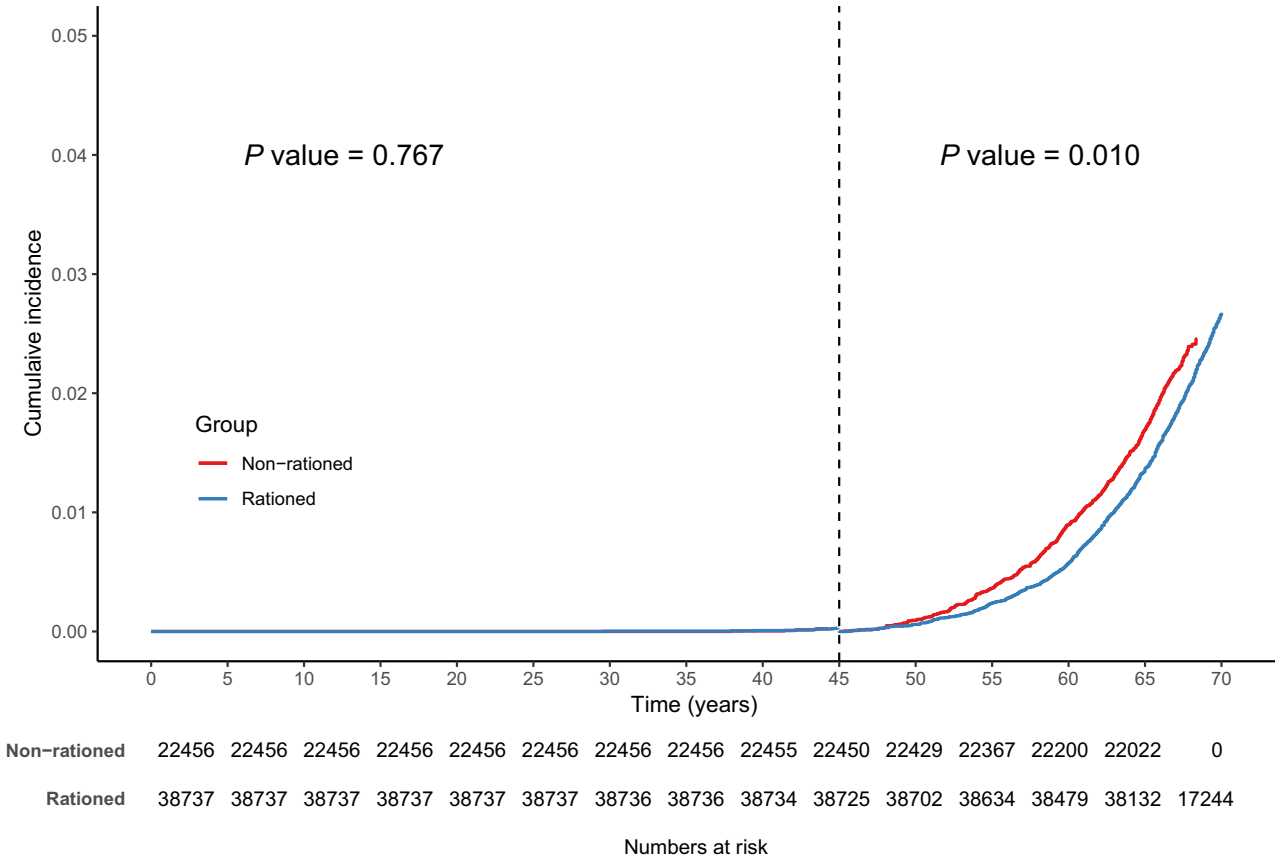

Fig. 2 | Kaplan–Meier estimates of incidence before and after the landmark time. The red line represents non-rationed participants and the blue line represents rationed participants. Incidence curves were estimated using the Kaplan–Meier method and compared using the log-rank test. All statistical tests were two-sided. No adjustment for multiple comparisons was applied. Source data are provided as a Source Data file.

induced by adaptive responses to maternal cues regarding her health or physical condition. Sugar rationing may have provided health benefits by altering the physiological metabolism in the uterus[16]. Moreover, excessive sugar consumption in early life may have affected the developing brain's sensitivity to sweetness, resulting in an increased preference for sugary food later in life[21].

This study, based on a natural experiment, explored the influence of early-life sugar rationing on the lifelong risk of HF. This protective effect persisted even to middle age. We proposed that early-life sugar management may have been beneficial for the prevention of HF, potentially yielding effects comparable to or superior to diabetes management and smoking cessation in the white population (given the similar PAFs), and this effect was markedly more pronounced than that of hyperlipidaemia management[22].

Several limitations should be acknowledged. First, the observational design of the study precluded the establishment of causal inferences. Consequently, our conclusions were preliminary, and caution should be exercised in extrapolating these findings to practical contexts. Although randomised controlled trials (RCTs) can theoretically establish causality, they are frequently impractical and ethically problematic in this context. Numerous diseases necessitate extended exposure durations to become evident. Moreover, assigning participants to potentially unhealthy dietary regimens for prolonged periods is unethical. Future research can implement a target trial emulation (TTE) framework as an alternative. Introduced in 2016 by Hernan[23], TTE seeks to utilise observational data to replicate a hypothetical randomised trial when an actual RCT is impractical. Methods such as cloning, censoring, and weighting are applied to synchronise time zero and reduce immortal time bias, in addition to propensity score methods and additional strategies to simulate randomisation[24]. Evidence indicates that when meticulously designed and assessed, studies employing the TTE framework can formulate conclusions comparable to those derived from randomised trials[25,26]. Future research should dynamically monitor maternal and early-life sugar consumption in a non-interventional framework and should be combined with a TTE study design, to yield ethically viable and practically pertinent causal evidence for examining the effects of early-life sugar rationing on subsequent health outcomes. Second, the evaluation of specific food or nutrients' effects on diseases should be independent of overall energy intake[27,28]. The average intake of other nutrients, including total energy, exhibited minimal variation before and after sugar rationing, and the predominant increase was attributable to sugar intake. This finding facilitates the identification of the independent effect of sugar rationing. However, owing to the natural experiment design of this study, we were unable to accurately measure individual-level variations in total energy intake. Third, although we adjusted for some time-invariant confounders, unmeasured differences may have persisted in individuals who were in utero or born during the sugar rationing period. We confined the analysis to a limited time frame and observe similar traits across cohorts. No significant changes in diets, except sugar consumption, were found before and after rationing. Furthermore, expanding upon the primary analysis using the birth cohort design, we performed a supplementary analysis employing the baseline assessment in the UKB as an alternative initiation point for follow-up. At this stage, participants had predominantly attained middle and advanced age. We further adjusted for their present socioeconomic and health conditions (e.g., body mass index [BMI], DM, hypertension and CVD), lifestyle factors (physical activity, smoking, alcohol consumption and dietary patterns), early-life growth and developmental factors. These analyses confirmed the robustness of our findings. Fourth, the UKB data lacked national representativeness, suggesting that the

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Table 2 | Association between sugar rationing and heart failure

	Crude model			Adjusted model		
	HR (95%CI)	*P* value	PAF (%)	HR (95%CI)	*P* value	PAF (%)
Cox proportional hazards model						
Rationing status						
Non-rationed	Reference			Reference		
Rationed	0.87 (0.78, 0.97)	0.011	4.45 (1.25–7.66)	0.87 (0.78, 0.97)	0.016	4.27 (1.03–7.50)
Duration of rationing						
Never rationed, born 21 months after rationing	Reference			Reference		
Never rationed, born 9–21 months after rationing	0.83 (0.70, 0.99)	0.039		0.84 (0.70, 0.997)	0.046	
Rationed in-utero only	0.79 (0.66, 0.95)	0.012		0.80 (0.66, 0.96)	0.017	
In-utero + rationed up to 12 months	0.78 (0.66, 0.92)	0.004		0.79 (0.66, 0.93)	0.005	
In-utero + rationed up to 24 months	0.77 (0.65, 0.91)	0.002		0.78 (0.66, 0.92)	0.004	
Trend test		0.006			0.008	
Flexible parametric survival models						
Rationing status						
Non-rationed	Reference			Reference		
Rationed	0.86 (0.77, 0.95)	0.005	4.84 (1.73–7.90)	0.86 (0.77, 0.96)	0.007	4.67 (1.53–7.80)
Duration of Rationing						
Never rationed, born 21 months after rationing	Reference			Reference		
Never rationed, born 9–21 months after rationing	0.83 (0.70, 0.99)	0.034		0.83 (0.70, 0.99)	0.040	
Rationed in-utero only	0.79 (0.65, 0.94)	0.009		0.79 (0.66, 0.95)	0.013	
In-utero + rationed up to 12 months	0.77 (0.65, 0.91)	0.002		0.78 (0.66, 0.92)	0.003	
In-utero + rationed up to 24 months	0.75 (0.64, 0.88)	0.001		0.76 (0.64, 0.89)	0.001	
Trend test		0.002			0.003	

CI confidence interval, *HR* hazard ratio, *PAF* population attributable fractions. The crude model did not account for covariates, while the adjusted model accounted for calendar month of birth, birthplace, sex, race, parental cardiovascular disease, parental hypertension, parental diabetes mellitus, and polygenic risk score of heart failure. Bold font is used solely to distinguish group headers and model types, with no statistical significance implied. All statistical tests were two-sided. No adjustment for multiple comparisons was applied.

Table 3 | Interaction between sugar rationing and genetic categories

	Low/intermedia PRS tertile HR (95%CI)	High PRS tertile HR (95%CI)
Rationed	Reference	1.19 (1.05, 1.35)
Non-rationed	1.06 (0.92, 1.22)	1.53 (1.31, 1.79)
Multiplicative scale	1.22 (0.98, 1.51), *P* value = 0.072	
RERI	0.29 (0.02, 0.55), *P* value = 0.019	
AP	0.19 (0.02, 0.35), *P* value = 0.012	

AP attributable proportion due to interaction, *CI* confidence interval, *HR* hazard ratio, *PAF* population attributable fractions, *PRS* polygenic risk score, *RERI* relative excess risk due to interaction. Model was adjusted for calendar month of birth, birthplace, sex, race, parental cardiovascular disease, parental hypertension, and parental diabetes mellitus. All statistical tests were two-sided. No adjustment for multiple comparisons was applied.

participants may have been more affluent and healthier. Consequently, subsequent research should integrate more representative, multi-regional data to improve the generalisability of the results.

In conclusion, limiting sugar intake during early life may promote a reduction in the risk of HF or delay of its onset. However, further studies that are feasible and ethically sound are needed to confirm these findings and to explore the underlying biological mechanisms.

Methods

Background

The UK National Food Survey (NFS), initiated in 1940[29], offers historical data on population dietary patterns. The survey initially targeted urban working-class households to assess wartime food policies. Beginning in 1950, it was broadened to encompass a nationally representative sample that included urban and rural regions, providing a more thorough insight into dietary habits. A stratified sampling design categorised households by residential area, social class and household composition to ensure representativeness. Surveys were administered quarterly, encompassing the initial two months of each quarter. Participating households documented all food purchases and consumption, excluding sweets, mineral water and alcoholic beverages. Food was quantified by weight or volume (e.g., grams and liters) and converted into per capita intake, energy content and essential nutrients (including protein, fat, carbohydrates and cereals) by using standardised food composition tables. Additional information can be found at https://www.gov.uk/government/statistics/family-food-historic-reports.

According to the UK NFS, per capita daily consumption of foods and nutrients, including protein, fat and produce, remained relatively stable before and after the sugar rationing period (1950 Q1–1953 Q2), while sugar intake significantly increased (Fig. 1a and Supplementary Fig. 2). Significantly, absolute variations in average individual energy intake were negligible, with relative changes of only 3%, approximately three-quarters of which were due to heightened sugar consumption. This proportion may be underestimated because the NFS omitted sweets and implemented a 10% adjustment to caloric intake for potential wastage and food acquired outside the home (Supplementary Figs. 3–5). Consequently, concentrating on a limited temporal framework with largely analogous cohort attributes and negligible dietary alterations aside from sugar consumption facilitates the discernment of the independent effects of sugar rationing on health outcomes.

Study design

This study employed data from the UKB, a prospective cohort study comprising over 500,000 participants from England, Scotland, and Wales, recruited between 2006 and 2010[30]. Socio-demographic,

medical and lifestyle data were collected using questionnaires, interviews and health records, and physical measurements and biological specimens were acquired using standardised protocols. The study was conducted in accordance with the Declaration of Helsinki. The North West Multicenter Research Ethics Committee approved the study, and all participants provided written informed consent. The study was performed under UKB application number 205837.

A birth cohort design was utilised, employing age as the temporal scale for follow-up[8]. Figure 1B illustrates that from the original 502,137 participants, we identified 74,196 individuals born between October 1951 and March 1956. We additionally excluded participants who were adopted, part of a multiple birth, pregnant at recruitment, born outside the UK (England, Wales or Scotland), lacked available HF follow-up data or had unavailable covariates, yielding a final study population of 61,193 participants. Figure 1C delineates the timeline of sugar rationing, categorising individuals into the rationed group ($N = 38{,}737$) and the non-rationed group ($N = 22{,}456$) according to their birth year and month. Moreover, participants were classified into five subgroups based on the timeline: in utero + rationed up to 24 months, in utero + rationed up to 12 months, rationed in utero only, never rationed + born 9–21 months after rationing and never rationed + born 21 months after rationing.

Follow-up of outcome

In the UKB, the initial occurrences of health-related outcomes for participants were ascertained by amalgamating self-reported data, primary care records, hospital admissions and death registry information. Health outcomes were classified using the International Classification of Diseases (10th Edition), and HF was coded as I50. The onset time of HF was determined by subtracting the birth date from the date of initial diagnosis. Follow-up outcomes were delineated as the earliest manifestation of HF, loss to follow-up or the end of the follow-up period (December 2022).

Covariates

This study primarily utilised a birth cohort design, focusing on the management of early-life risk factors. The exposure period to sugar rationing was confined to the initial 1000 days of life, encompassing both infancy and the prenatal stage. Currently, conventional risk factors for HF, such as smoking, physical activity, BMI, hypertension and diabetes history, are either unquantifiable or lack clinical significance and comparability. Consequently, drawing from prior research[8], the primary analysis considered the following time-invariant covariates: birthplace, sex, race, parental CVD, parental hypertension, parental DM, calendar month of birth, and PRS for HF.

The birthplace was classified as England, Wales or Scotland. Sex was classified as male and female. Race was categorised as White and other classifications. Parental CVD was defined as the participant indicating that either their father or mother had experienced heart disease or stroke. Parental hypertension and DM were also self-reported. Calendar month of birth indicators (with January as the reference category) were incorporated to account for seasonality. The PRS for HF was computed using summary statistics from the genome-wide association study conducted by Shah et al.[31] and PRSice-2, which calculates PRS across a range of thresholds to determine the optimal one for explaining phenotypic variance (R^2)[32].

Statistical analysis

Baseline characteristics were classified according to exposure to sugar rationing. Continuous variables were expressed as means ± SDs, whereas categorical variables were presented as numbers and percentages (%). Group differences for continuous and categorical variables were assessed utilising the *t*-test and Pearson's Chi-squared test, respectively. Cohen's *d* and Cramér's *V* were computed to assess the magnitude of differences between groups[33].

The cumulative incidence of HF was determined for participants exposed to sugar rationing compared to those not exposed, utilising the Kaplan–Meier method, with differences evaluated via the log-rank test. Landmark analysis was conducted by establishing an alternative starting point (landmark) at the curve's inflection point (45 years)[34]. Two models were used to assess the association between the status and duration of sugar rationing and the risk of HF over the life course. The initial method employed was traditional Cox regression, which posits that the influence of exposure variables on the outcome remains constant over time, referred to as the proportional hazards assumption. The second method utilised FPSM, enabling the effects of exposure factors to fluctuate over time. These models employ penalised splines to refine the hazard function, adeptly encapsulating intricate, nonlinear associations[35,36]. The crude model did not account for covariates, whereas the adjusted model accounted for the calendar month of birth, birthplace, sex, race, parental CVD, parental hypertension, parental DM, and PRS for HF. Furthermore, PAFs were calculated using Miettinen's formula[37,38], where $PAF = P_c\ (R-1)/R$, P_c represents the proportion of HF cases unexposed to sugar rationing, and R is the estimated HR of HF for the unexposed versus the exposed group. The 95% CIs were calculated using the delta method[39].

We further subdivided the duration of rationing into six-month intervals and used linear regression to examine the association between the status and duration of sugar rationing and the age of HF onset. Additionally, the interaction between genetic susceptibility and sugar rationing on HF risk was evaluated. To assess the multiplicative interaction, we incorporated an interaction term between the two factors in regression model and computed the RERI and AP to evaluate the additive interaction[40,41].

The robustness of our findings was assessed using two sensitivity analyses. First, we reexamined the association between sugar rationing and HF with a parametric proportional hazards model based on the Gompertz distribution, which assumes that the hazard increases exponentially over time. Second, we discarded the initial birth cohort design, which commenced follow-up at birth, and opted to utilise the baseline assessment of participants in the UKB (2006–2010) as the follow-up initiation point. At this stage, most participants were in midlife, and a small subset had encountered outcome events (loss to follow-up or incident HF, $n = 187$), which were excluded from the analysis. Four categories of potential confounders were considered: (1) time-invariant covariates (birthplace, sex, race, parental CVD, parental hypertension, parental DM, calendar month of birth, and PRS for HF); (2) current socioeconomic and health status, including BMI, DM, hypertension, and CVD; (3) lifestyle factors, including physical activity, smoking, alcohol consumption, and dietary patterns; and (4) early-life growth and developmental factors. Socioeconomic status was assessed using the Townsend deprivation index at recruitment, which incorporated data on social class, employment, car availability, and housing. BMI was calculated by dividing weight in kilograms by the square of height in meters. DM history was characterised by self-reported physician-diagnosed DM, current use of antidiabetic medication or insulin, fasting plasma glucose levels of ≥7.0 mmol/L or glycated hemoglobin levels of ≥6.5% (48 mmol/mol). Hypertension history was characterised by self-reported physician-diagnosed hypertension, current use of antihypertensive medication or repeated measurements indicating an average systolic blood pressure of ≥140 mmHg or diastolic blood pressure of ≥90 mmHg. CVD history was defined as self-reported physician-diagnosed heart attack, angina, or stroke. Physical activity was categorised into walking, moderate and vigorous activity according to total weekly metabolic equivalent task hours for all activities. Smoking and alcohol consumption were categorised as never, former or current. The evaluation of a healthy dietary pattern was conducted according to the consumption of fruits, vegetables, whole grains, processed meat, refined grains, fish, unprocessed red meat and sugar-sweetened food or SSBs. Scoring details are

provided in Supplementary Table 5[42]. Early-life growth and development factors encompassed self-assessed body size and height at age 10 relative to peers (thinner/shorter, about average, plumper/taller). To minimise the influence of the birth cohort effect, we performed survival regression analyses, using age as the temporal scale to investigate the association between early-life exposure to sugar rationing and the subsequent risk of HF in midlife and later life. We excluded participants who encountered outcomes during the initial two years of follow-up were subsequently excluded to reduce potential reverse causation.

All analyses were conducted using R (version 4.4.1), PLINK (version 2.0), and Free Statistics software (version 2.0). Statistical tests were two-sided, and significance was set at $P < 0.05$.

Ethics statement
The UK Biobank was approved by the North West Multicenter Research Ethics Committee, with all participants providing written informed consent.

Ethical approval and informed consent
Were waived as the UK Biobank data is publicly available and does not include identifiable information.

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
The raw data supporting the findings of this study are available from the UK Biobank (https://www.ukbiobank.ac.uk/) under restricted access due to data privacy and ethical restrictions. Access to these data can be obtained by submitting an application directly to the UK Biobank. This research was conducted under UK Biobank application number 205837. Source data are provided with this paper.

Code availability
The code used in this study is openly available at: https://github.com/losidan/Code.

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Author contributions

Conceptualization: H.T., X.Z., Y.C.; Data Management and Analysis: H.T., X.Z., JT.H.; Figure Creation: H.T., JN.H., H.L.; Writing - Original Draft Preparation: H.T., X.Z., X.C., C.T., N.L., M.L.; Writing - Review and Editing: X.C., Q.Y., S.W., P.C., J.W., L.J., Y.Z.; Provided Critical Revisions to the Manuscript: H.T., X.Z., JT.H., Y.W., X.T., Y.C.; Project Management: Y.W., X.T., Y.C.

Competing interests

The authors declare no competing interests.

Additional information

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